ALEXANDER MEJÍA OSPINA

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PROFESSIONAL SUMMARY

Seventh-semester Software Engineering student who builds and ships complete projects: a deployed Java and Spring Boot REST API with automated tests, a Python and SQL data pipeline over 1.18 million public health cases, a Flutter mobile app, and a React website in production. Available now for a full-time internship as a software development intern or data analyst.

EDUCATION

Universidad Cooperativa de Colombia — Medellín, Colombia
B.S. in Software Engineering — 7th semester — 2023 – Present
Relevant Coursework: Data Analytics, Software Testing, Software Implementation and Maintenance, Numerical Methods.
Scholarships: Recipient of the Sapiencia – Participatory Budget scholarship (Medellín).

Colegio Parroquial San Buenaventura — Medellín, Colombia
High School Diploma (Academic Track) — 2012 – 2022

TECHNICAL SKILLS

Languages: Java, Python, Dart, JavaScript, C++, SQL, HTML, CSS
Backend: Spring Boot, Spring Data JPA, REST APIs, Swagger/OpenAPI, FastAPI, Flask, Maven
Data Analysis: SQL (CTEs, window functions), Python, Pandas, Power BI, Looker Studio, Jupyter, Google Colab, linear and logistic regression, ANOVA
Web and Mobile: HTML, CSS, Bootstrap, JavaScript, React, Flutter, Riverpod, MVVM architecture
Testing and DevOps: JUnit, Mockito, pytest, Postman, Docker, GitHub Actions (CI), Git, GitHub
Databases and Cloud: MySQL, SQLite, Firebase, Google Cloud Platform, Render
Methodologies: Scrum, software testing (test lead on an academic project) | **Spoken Languages:** Spanish (native), English (intermediate, B1)

PROJECTS

Antioquia Public Health Observatory (live · code) — *Personal project* — 2026
- Built a Python and Pandas ETL pipeline that downloads 1.18 million cases reported to Colombia's SIVIGILA surveillance system (2007–2022) from open data, cleans and validates them, and loads them into a MySQL star schema.
- Wrote SQL queries with CTEs and window functions for trends, municipal rankings, and endemic channels; published the findings on a website with charts and a FastAPI API.
- Automated pytest tests and monthly data refreshes with GitHub Actions.

CineX — Cinema Management System (live · code) — *v1.0 team, v2.0 solo* — 2025 – 2026
- v1.0 (team): set up the repository, integrated the team's work, and reviewed pull requests for a Java 21 Spring Boot REST API and its web admin panel.
- v2.0 (solo): moved movies, rooms, showtimes, and bookings to MySQL with Spring Data JPA, adding business rules (room capacity, no overlapping showtimes), validation, and JSON errors documented with Swagger.
- Added 17 JUnit and Mockito tests, a 26-request Postman collection, Docker Compose, and GitHub Actions CI; the API is deployed on Render with MySQL on TiDB Cloud.

Numi — Offline Educational App with AI Assistant — *Team project* — 2026
- Flutter app for elementary students in rural Medellín that works without internet, with 15 activities across 5 subjects; MVVM architecture with Riverpod and progress synced to a FastAPI backend.
- My role: designed the app architecture, built the logic and screens for Natural Sciences, the offline RAG for the AI assistant (on-device Gemma 2B with a SQLite knowledge base), and the responsive design.

Casa Rendón — Website (casarendon.com) — *Team project, in production* — 2026
- Designed and built the website of an artisanal coffee and wine brand: React components for the catalog, gallery, and FAQ, and a visit-booking form that sends requests via WhatsApp.

Data Analysis with Python — *Data Analytics course* — Oct. 2026
- Linear regression and ANOVA to predict fuel efficiency (Auto MPG) and logistic regression to classify benign and malignant tumors, using Pandas in Jupyter.

ADDITIONAL TRAINING

Java and JavaScript Programming Certificate Programs — Politécnico de Colombia — Completed
C++ Essentials 1 — Cisco Networking Academy and OpenEDG C++ Institute — Oct. 2025
Google Cloud — Badges: App Dev Environment, Cloud Load Balancing for Compute Engine, and Cloud Computing Foundations (2) — 2026

REFERENCES

Available upon request.